Syllogistic Entailment over Distributional Vectors

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M781/L715

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Alrighty, so for my project I'm maybe doing too much – I'm trying to do syllogistic entailment over distributional vectors.

You can probably guess from class what 'syllogistic entailment' means – entailment with words like 'all', 'some', 'no'. But probably not all of you know what 'distributional vectors' means...

A Crash Course in Distributional Semantics

... so I'll begin with a crash course in distributional semantics.

Before I get into it, I should say that most of this I'm pulling from a fantastic read called Frege in Space. I'll cite this and everything else I'm pulling from at the end. If you'd like to track something down, please ask and I can hook you up.

Traditional semantics

- Has been tremendously successful

The kind of semantics that we've been doing in this class is powerful; we've seen how it offers rich explanations for how different phrases work and how they combine. In this class alone, we've given reasonable denotations to nouns, noun phrases, relations, some quantifiers, transitive verbs, and reflexive verbs.

Despite this, traditional semantics suffers from some drawbacks:

- Coming up with the semantics of a word requires careful analysis of a word – and this takes a lot of time. So far, we've barely scratched the surface. Semantics hasn't touched on most abstract, uncommon words (e.g. 'nauseant', 'bravery').
- In addition, we tend to make simplifying assumptions about how words are used. The kinds of sentences that linguists consider are very 'sterile' compared to real sentences that writers and speakers of a language write and speak. (My favorite: "John, Bill and Tom killed each other," from The Phrasal Lexicon)
- Languages have a lot of words! It's just not feasible to do a by-hand careful analysis of every use of every word in a language (like we did for 'all', 'sees', 'most', 'some', ...)

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- There are *a lot* of words to analyze!

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Meaning from real text

Ultimately, we would like to be able to assign meaning to any word, phrase, or sentence. We could imagine doing this in a data-driven way: have a program pour over real text, and extract the meanings of words 'automatically'.

How can we do this? The crux is what's called the distributional hypothesis: [show]

The often-quoted tagline is that "words occuring in similar linguistic contexts have similar semantic meaning." So in order to extract meaning from text, we only need to be able to extract occurrences from text (and text is full of occurrences).

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Meaning from real text

... of anticipating being hurt by another	horse	especially if some other rider comes ...
... was simply beaten by a better	horse	at the distance on the day ...
... can infer from these studies that	horses	reared with other horses in a ...
... studies that horses reared with other	horses	in a free and enriched environment ...
... people saying "Is that all your	horse	gets to eat?" in amazement. The ...
... and a cache of cattle and	horse	bones, while from Normangate Field a ...
... Bachelor's Button was a sterling good	horse,	especially at Ascot, but he was ...
... the same way as a domestic	horse	that it may be stabled with ...
... Attachment in 1790 – that is, one	horse	or two cows for each £4 ...
... grey hair as coarse as a	horse	's tail straying from her mob-cap ...

Here's an example from Guy Emerson's dissertation. Say we're pulling from a text that has the word 'horse' several times. The distributional hypothesis says that the meaning of the word 'horse' can be gleaned by looking at its contexts – the words it's surrounded by.

The word 'cow' has a similar meaning to the word 'horse' because it might occur in similar kinds of contexts. e.g. we might have sentences stating that cows are reared with horses, that they also eat, that they can also be domestic. But there are certain contexts that 'cow' can't share, for example that last sentence – a cow's tail hair isn't coarse like a horse's – and so the meaning of the two words differ a bit.

[next bit]

We want to store these contexts as a proxy for 'horse's meaning. The usual way to do this is to keep a vector for horse (think of it like an ordered listing). Each item in this listing is a function of the number of times 'horse' occurs in proximity to another given word in the corpus. ('In proximity' usually means 'within a window', like these sentence fragments with 'horse'.)

So each word gets a vector like this, and that vector is a proxy for the word's meaning. Usually these vectors are reduced to a smaller dimension, so that components represent less sensitive aspects of the word's context.

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$$\vec{v}_{\text{horse}} = \langle \underset{\text{Button}}{9} \quad \underset{\text{reared}}{4} \quad \underset{\text{rider}}{33} \quad \underset{\text{especially}}{1} \quad \ldots \rangle$$

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What we gain

- A method to automatically extract “meaning” of words straight from real text …

This approach to semantics offers an automated way to get proxies for word meanings without all of the tedious labor that we had to do in traditional semantics. Additionally, it works on real text, rather than the sterilized text that’s needed to come up with reasonable denotations.

To boot, this view also comes with a ready-made way to compare the similarity or closeness of words – we just compare the distance between their vectors!

But I should emphasize at this point that these vector representations work best for content words; nouns, adjectives and verbs are best described by their uses. This does not work so well with more logical words (like quantifiers and connectives).

For example, the meaning of the word ’and’ has almost nothing to do with any two phrases that might sit next to it! Like, the meaning of ’and’ here has nothing to do with how reliable or hardworking this person is.

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What we lose

And what about sentences? Should the meaning of a sentence also be a vector, based on occurrences of nearby sentences? Or maybe the meaning of a sentence should be built up from the meaning of its parts, just like in traditional semantics.

We could work along these lines – a lot of early research in distributional semantics was about which vector mixtures give better meanings for longer phrases. The issue with this idea is mainly that vector mixtures just treat the words they are combining as independent of each other – whereas we've seen that the meaning of complex sentences comes from the relation between words!

So we admit, for now, that we lose out on compositionality: This idea that the meaning of an expression is determined by the meaning of its parts.

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What we lose

COMPOSITIONALITY

- Explains human ability to understand novel sentences

This is such a fundamental feature of language that it's unthinkable to have a theory of semantics that doesn't account for this: First, it allows us to understand sentences we've never seen before. And second we need compositionality to do entailment. That is, entailment based on deduction rules (like we've been doing on class) relies on syntactically picking apart sentences, and this relies on compositionality.

This second point is important for us, since we would like to do entailment for this kind of semantics. We'll come back to this in a bit.

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Compositional distributional semantics

In response to the compositionality issue, there have been lots and lots of proposals to fix distributional semantics by making it compositional.

Compositional distributional semantics

Bob Coecke, Mehrnoosh Sadrzadeh, and Stephen Clark. Mathematical foundations for a compositional distributional model of meaning. *Linguistic Analysis*, 36, A Festschrift for Joachim Lambek:345–384. 2010. Cited on pages 19 and 47.

Edward Grefenstette and Mehrnoosh Sadrzadeh. Experimental support for a categorical compositional distributional model of meaning. In *Proceedings of the 16th Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pages 1394–1404. Association for Computational Linguistics, 2011. Cited on page 114.

Richard Socher, Brody Huval, Christopher D. Manning, and Andrew Y. Ng. Semantic compositionality through recursive matrix-vector spaces. In *Proceedings of the 2012 Joint Conference on Empirical Methods in Natural Language Processing (EMNLP) and Computational Natural Language Learning (CoNLL)*, pages 1201–1211. Association for Computational Linguistics, 2012. Cited on page 38.

Esma Balkır. *Using density matrices in a compositional distributional model of meaning*. Master's thesis, University of Oxford, 2014. Cited on pages 32, 34, and 36.

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Denis Paperno, Nghia The Pham, and Marco Baroni. A practical and linguistically-motivated approach to compositional distributional semantics. In *Proceedings of the 52nd Annual Meeting of the Association for Computational Linguistics (ACL), Long Papers*, pages 90–99, 2014. Cited on page 45.

Tamara Polajnar, Laura Rimell, and Stephen Clark. An exploration of discourse-based sentence spaces for compositional distributional semantics. In *Proceedings of the EMNLP Workshop on Linking Models of Lexical, Sentential and Discourse-level Semantics (LSDSem)*, pages 1–11. Association for Computational Linguistics, 2015. Cited on page 38.

Stephen Clark, Laura Rimell, Tamara Polajnar, and Jean Maillard. The categorial framework for compositional distributional semantics. Unpublished draft, 2016. Cited on page 48.

I won't get into the details here. I just want to make the impression that there has been a ton of work on this issue.

Compositional distributional semantics

We mentioned earlier that one big reason we want compositionality is for doing entailment. This quote, pulled straight from Guy Emerson's Functional Distributional Semantics (which I'll cite at the end) points out just that: None of these proposals give an account of inference, despite accounting for compositionality.

In fact, giving an appropriate definition of entailment for sentences built from distributional vectors is a big open problem! At this stage, any system whatsoever that can do inference in this way would be considered a big step forward.

Compositional distributional semantics

Current approaches to distributional semantics generally involve representing words as points in a high-dimensional vector space. However, vectors do not provide 'natural' composition operations that have clear analogues with operations in formal semantics, which makes it challenging to perform inference, or capture various aspects of meaning studied by semanticists. This is true whether the vectors are constructed using a *count* approach (e.g. Turney and Pantel, 2010) or an *embedding* approach (e.g. Mikolov et al., 2013), and indeed Levy and Goldberg (2014b) showed that there are close links between them. Even the tensorial approach described by Coecke et al. (2010) and Baroni et al. (2014), which naturally captures argument structure, does not allow an obvious account of context dependence, or logical inference.

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The Entailment Problem

That's not to say there's been no work done on this already, but there hasn't been much. First, the vast majority of work done in entailment focuses on lexical entailment – i.e. relationships between individual words. That's not the problem at stake here – we want to define entailment between full sentences.

After a good amount of searching, this was all that turned up for full-sentence entailment:

[show papers]

The first one extends lexical entailment to two-word phrases, like noun-phrases and quantifier phrases. But this paper doesn't address entailment beyond this point at all.

The next two I can't really say much about, since I don't completely understand the tensorial framework suggested by Coecke. But despite doing entailment, they leave open the interpretation of logical words (like connectives and quantifiers), which to my mind are necessary to do any interesting inference.

The last one is fun: It uses probabilistic programming semantics to give a formal account of bayesian inference between two sentences. Unfortunately, this paper is very "throw the kitchen sink at it," and so it's hard for me to glean any specific insight from it.

What still seems to be open is doing interesting inference between full sentences, with connectives or quantifiers, in a principled way.

The Entailment Problem

Marco Baroni et al. "Entailment above the word level in distributional semantics". In: *Proceedings of the 13th Conference of the European Chapter of the Association for Computational Linguistics*. 2012, pp. 23–32

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Jean-Philippe Bernardy et al. “Bayesian inference semantics: A modelling system and a test suite”. In: *Proceedings of the Eighth Joint Conference on Lexical and Computational Semantics (* SEM 2019)*. 2019, pp. 263–272

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Plan of Attack

This is a tall order – and I don't plan on reconstructing all of semantics by Christmas!

So we need to start small. The idea is that we can start with a 'toy' semantics to play around with, and hopefully the insights we gain will carry over in the more complicated setting.

Larry did the same thing when teaching us semantics. And you all remember what we started with...

Syllogistic logic

All a b

Some a b

No a b

Syllogisms! We started this class by studying the meaning of All. We've also talked (if I'm remembering right) about Some and .

[show semantics] The semantics for these relational quantifiers is straightforward: 'All' means subset, 'Some a are b' means that the intersection is not empty, and 'No a are b' means that the intersection is empty.

I'm going to focus on 'All a are b', since this is the one you all are most familiar with.

Syllogistic logic

All a b	denotes	$[\![a]\!] \subseteq [\![b]\!]$
Some a b	denotes	$[\![a]\!] \cap [\![b]\!] \neq \varnothing$
No a b	denotes	$[\![a]\!] \cap [\![b]\!] = \varnothing$

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The set/individual distinction

$$\mathcal{M} \models \mathsf{All}\ a\ b \quad \text{iff} \quad [\![a]\!] \subseteq [\![b]\!]$$

So a model models All a b again means that a is a subset of b. This means that every individual contained in set a is also contained in set b.

So in order to assign a meaning to All, we at least have to make a distinction between 'set' nouns (like 'woman') and 'proper'/individual nouns (like 'Sarah').

Distributional semantics doesn't make this distinction by default – all words are assigned the same vector object. So we're going to have to make this distinction ourselves.

One thing I should point out is that in distributional semantics, our individuals should come from the text. The problem here is that not all nouns have explicit 'individual' proper nouns lying around. For example, we may have 'Indiana' as an entity inside 'states', but what (if anything) is an entity of 'tree'?

I haven't figured out how to fix this problem yet, and so for now I'm going to assume that a and b have at least one corresponding proper noun in the text.

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A working analogy

Frege	Distributional
individuals	vectors
sets	vector spaces

We can draw an analogy with traditional semantics: In the tradition of Frege, proper nouns are typed 'individuals,' and nouns are typed 'sets' of individuals. Similarly, if our proper nouns are assigned occurrence vectors, we can assign to our nouns spaces of nearby vectors.

Because we're working within distributional semantics, we want this to work in a data-driven way; we don't want to manually assign a vector space to each noun. So we need some automatic way to go from a vector to a vector space.

Lifting content words to spaces

$$[\![\textit{horse}]\!] = \{\vec{v}_i \mid i \text{ is an individual and } \vec{v}_i \cdot \vec{v}_{\text{horse}} > \Theta_{\text{horse}}\}$$

I'm going to call this 'lifting' – we are raising the vector to a higher type. The way I do this I'm borrowing from the Bayesian Inference Semantics paper I mentioned earlier.

Let's use 'horse' as an example again. The trick is that we can say that another vector is close to the 'horse' vector if their dot product ($\vec{v}_i$ projected onto horse) exceeds some special threshold for the word 'horse'. And if we restrict the kinds of vectors we're looking at to individuals, then we get the ball of nearby individual vectors. And my hope is that this gives us the individuals contained in 'horse'.

I should mention that this doesn't work at all if we don't restrict i to individuals; the words 'horse' and 'cow' might be very similar to one another, but there's no way the word 'cow' is contained in 'horse'.

Back to All

$$\mathcal{M} \models \text{All } a\ b \quad \text{iff} \quad [\![a]\!] \text{ is a subspace of } [\![b]\!]$$

With this 'vector lifting', the straightforward way to redefine All is as subspace rather than subset. But then this means that "every individual word that is a-like is also b-like".

This now has a typicality reading: All typical a's are typical b's, and this degree of 'typical' depends on our choice of threshold for a and b.

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So that's the basic idea. Here's a short list of what I have left, that I'd like to do by mid-December.

My Todo List

Playing in the sandbox

- Play with space of all x-like vectors in word2vec

First, I want to get a full working-example of this vector-lifting in word2vec. I've been playing around with word2vec a little already. I'm not super confident that just getting the space of nearby vectors will give me, say, the horses contained in 'horse', so I need to play with other options.

I would also like to tweak the thresholds for each word – how small does the threshold have to be, for example, to restrict our individuals to exactly 'horse' individuals?

The next thing to take on is a pretty big deal: What do we do in cases where our nouns a and b have no corresponding proper nouns? How do we infer that all 'cakes' are 'edible'? I don't have a good answer to this, though I'm thinking of ways to make other words act as 'surrogate' individuals.

Finally, since this is a distributional system, we should test it on real text. I hope that it makes mostly correct inferences. I'm not sure what suite to test it on, though, because the syntax is so restricted that it can't really handle most datasets. If anybody has any suggestions, I'm open.

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Playing in the sandbox

- Play with space of all x-like vectors in word2vec
- Play with thresholds Θ_w
- What do we do in cases with *no* proper nouns?
- Comparing this entailment to other proposals

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Remaining references

I couldn't 'pick a slide' to cite these because I relied on them all throughout the slides.

Marco Baroni, Raffaella Bernardi, Roberto Zamparelli, et al. "Frege in space: A program for compositional distributional semantics". In: *Linguistic Issues in language technology* 9.6 (2014), pp. 5–110

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